

(a) Actual setup (b) Circuit diagram

Figure 4.6 Ammeter in a circuit

Problem 2

If, 25 C of charge is determined to pass through a wire of any cross section in 50 s, what is the measure of current?

Solution:

I = q / t = (25 C) / (50 s) = 0.5 C/s = 0.5 A

Problem 3

The current flowing through a lamp is 0.2A. If the lamp is switched on for one hour, what is the total electric charge that passes through the lamp?

Solution:

I = q / t; q = I x t

1hr = 1 × 60 × 60 s = 3600 s

q = I × t = 0.2 A × 3600 s = 720 C

4.2.3 Electromotive force (e.m.f)

Imagine that two ends of a water pipe filled with water are connected. Although filled with water, the water will not move or circle around the tube on its own. Suppose, you insert a pump in between and the pump pushes the water, then the water will start moving in the tube. Now the moving water can be used to produce some work. We can insert a water wheel in between the flow and make it to rotate and further use that rotation to operate machinery.

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Likewise if you take a circular copper wire, it is full of free electrons. However, they are not moving in a particular direction. You need some force to push the electrons to move in a direction. The water pump and a battery are compared in Figure 4.7.

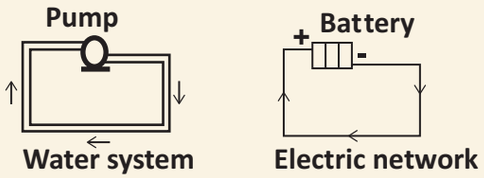


Figure 4.7 Battery is analogues to water pump

Devices like electric cells and other electrical energy sources act like pump, 'pushing' the charges to flow through a wire or conductor. The 'pumping' action of the electrical energy source is made possible by the 'electromotive force' (e.m.f). The electromotive force is represented as (ε). The e.m.f of an electrical energy source is the work done (W) by the source in driving a unit charge (q) around the complete circuit.

$\varepsilon = W/q$ where, W is the work done. The SI unit of e.m.f is joules per coulomb (JC^{-1}) or volt (V). In other words **the e.m.f of an electrical energy source is one volt if one joule of work is done by the source to drive one coulomb of charge completely around the circuit.**

Problem 4

The e.m.f of a cell is 1.5 V. What is the energy provided by the cell to drive 0.5 C of charge around the circuit?

Solution:

ε = 1.5 V and q = 0.5 C

ε = W/q; W = ε × q;

Therefore W = 1.5 × 0.5 = 0.75 J

4.2.4 Potential difference (pd)

One does not just let the circuit connect one terminal of a cell to another. Often we connect, say a bulb or a small fan or any other electrical device in an electric circuit and use the electric current to drive them. This is how a certain amount of electrical energy provided by the cell or any other

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